Mathematical Workshop

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Introduction

Mathematical Workshop is an evolving collection of mathematical problems and solutions written in the spirit of mathematical competitions and classical problem solving.

The goal of this project is to gather and preserve the problems that I solve and publish over time on my personal blog:

https://evenson0.github.io

Rather than letting each problem remain isolated in a blog post, this book collects them into a single workshop of ideas where arguments, techniques, and observations can be refined and organized.

The problems presented here cover various areas of elementary mathematics, especially combinatorics, algebra, and number theory. Each problem is accompanied by a concise solution emphasizing the key mathematical idea.

The collection is intended to grow gradually over time as new problems are solved and published.

Problem 1. *Find all integer solutions* $(a, b, c) \in \mathbb{Z}^3$ *satisfying*

$$a^2 + b^2 + c^2 = a^2 b^2.$$

(Number Theory, Diophantine Equations, Modular Arithmetic, Parity)

Solution. *We prove that the only integer solution is*

$$(a, b, c) = (0, 0, 0).$$

Assume that (a, b, c) *is a solution and that at least one of* a, b, c *is non-zero.*
Let 2^k *be the highest power of* 2 *dividing all three integers* a, b, c*. Write*

$$a = 2^k a_1, \qquad b = 2^k b_1, \qquad c = 2^k c_1,$$

where a_1, b_1, c_1 *are integers and not all even.*

Remark. *Since* 2^k *is defined as the largest power of* 2 *dividing* a, b, c*, the integers* a_1, b_1, c_1 *cannot all be even. Indeed, if they were all even, we could write*

$$a_1 = 2a_2, \qquad b_1 = 2b_2, \qquad c_1 = 2c_2,$$

which would imply

$$a = 2^{k+1} a_2, \qquad b = 2^{k+1} b_2, \qquad c = 2^{k+1} c_2.$$

But then 2^{k+1} *would divide* a, b, c*, contradicting the maximality of* k*. Therefore at least one of* a_1, b_1, c_1 *must be odd.*

Substituting into the equation gives

$$(2^k a_1)^2 + (2^k b_1)^2 + (2^k c_1)^2 = (2^k a_1)^2 (2^k b_1)^2.$$

Hence

$$2^{2k}(a_1^2 + b_1^2 + c_1^2) = 2^{4k} a_1^2 b_1^2.$$

Dividing by 2^{2k} *yields*

$$a_1^2 + b_1^2 + c_1^2 = 2^{2k} a_1^2 b_1^2. \tag{1}$$

Step 1 — Work modulo 4
Squares are congruent to 0 *or* 1 *modulo* 4*.*
Since at least one of a_1, b_1, c_1 *is odd, the left-hand side of (1) is congruent to*

$$1, 2, \text{ or } 3 \pmod 4.$$

Remark. *If an integer is odd, its square is congruent to* $1 \pmod 4$*, while the square of an even integer is congruent to* $0 \pmod 4$*. Since at least one of* a_1, b_1, c_1 *is odd, at least one of the squares* a_1^2, b_1^2, c_1^2 *is* $1 \pmod 4$*. Consequently the sum* $a_1^2 + b_1^2 + c_1^2$ *cannot be* $0 \pmod 4$*; it must be* $1, 2,$ *or* $3 \pmod 4$*.*

Now consider the right-hand side.
If $k \geq 1$ then 2^{2k} is divisible by 4, so

$$2^{2k} a_1^2 b_1^2 \equiv 0 \pmod 4.$$

But the left-hand side cannot be $0 \pmod 4$, giving a contradiction. Therefore

$$k = 0.$$

Equation (1) becomes

$$a_1^2 + b_1^2 + c_1^2 = a_1^2 b_1^2. \tag{2}$$

Step 2 — Force a_1 and b_1 to be odd
If either a_1 or b_1 were even, then $a_1^2 b_1^2$ would be divisible by 4, so the right-hand side of (2) would be

$$0 \pmod 4.$$

But the left-hand side contains at least one odd square, so it is congruent to

$$1, 2, \text{ or } 3 \pmod 4,$$

which is impossible.
Thus both a_1 and b_1 must be odd, and therefore

$$a_1^2 b_1^2 \equiv 1 \pmod 4.$$

Step 3 — Analyze c_1
Since a_1 and b_1 are odd,

$$a_1^2 \equiv b_1^2 \equiv 1 \pmod 4.$$

Hence

$$a_1^2 + b_1^2 + c_1^2 \equiv 1 + 1 + c_1^2 \equiv 2 + c_1^2 \pmod 4.$$

If c_1 is even, then $c_1^2 \equiv 0$ and

$$a_1^2 + b_1^2 + c_1^2 \equiv 2 \pmod 4.$$

If c_1 is odd, then $c_1^2 \equiv 1$ and

$$a_1^2 + b_1^2 + c_1^2 \equiv 3 \pmod 4.$$

In both cases the left-hand side is 2 or $3 \pmod 4$, whereas the right-hand side is $1 \pmod 4$.
This contradiction shows that our assumption was false.
Therefore the only integer solution is

$$\boxed{(a, b, c) = (0, 0, 0)}.$$

Problem 2. *In a round-robin tournament with n players $P_1, P_2, \ldots, P_n$, each player plays exactly one match against every other player.*

There are no ties, so every match results in one win and one loss.

Let W_k denote the number of wins of player P_k, and let L_k denote the number of losses of player P_k.

Show that

$$\sum_{k=1}^{n} W_k^2 = \sum_{k=1}^{n} L_k^2.$$

(Combinatorics, Graph Theory)

Solution. *The identity can be written*

$$\sum_{k=1}^{n} (W_k^2 - L_k^2) = 0$$

or

$$\sum_{k=1}^{n} (W_k - L_k)(W_k + L_k) = 0.$$

For each player we have

$$W_k + L_k = n - 1,$$

since each player plays against the other $n-1$ players.

Thus

$$(n-1) \sum_{k=1}^{n} (W_k - L_k) = 0.$$

Since $n - 1 \neq 0$, we must have

$$\sum_{k=1}^{n} (W_k - L_k) = 0.$$

Equivalently

$$\sum_{k=1}^{n} W_k = \sum_{k=1}^{n} L_k.$$

Every match produces exactly one win and one loss, so the total number of wins equals the total number of losses.

Therefore

$$\boxed{\sum_{k=1}^{n} W_k^2 = \sum_{k=1}^{n} L_k^2}.$$

Problem 3. *Let $P(x)$ be a polynomial of degree n satisfying*

$$P(k) = k \quad \text{for } k = 1, 2, \ldots, n$$

and

$$P(0) = 1.$$

Find

$$P(-1).$$

(Algebra, Polynomials)

Solution. *Define*

$$Q(x) = P(x) - x.$$

Then $Q(x)$ has roots

$$1, 2, \ldots, n.$$

Hence

$$Q(x) = c \prod_{k=1}^{n} (x - k)$$

for some constant c.
Thus

$$P(x) = x + c \prod_{k=1}^{n} (x - k).$$

Using $P(0) = 1$ gives

$$1 = c(-1)^n n!$$

so

$$c = \frac{(-1)^n}{n!}.$$

Therefore

$$P(-1) = -1 + \frac{(-1)^n}{n!} \prod_{k=1}^{n} (-1 - k).$$

But

$$\prod_{k=1}^{n} (-1 - k) = (-1)^n (n + 1)!.$$

Thus

$$P(-1) = -1 + (n + 1) = n.$$

Hence

$$\boxed{P(-1) = n}.$$

Problem 4. *Find the smallest positive integer n such that every digit of $15n$ is either 8 or 0.*

(Number Theory, Divisibility)

Solution. *Let*

$$N = 15n.$$

Since N is divisible by 15, it must be divisible by both 3 and 5.

Because N is divisible by 5, its last digit must be 0 or 5. The digits are restricted to 0 and 8, so the last digit must be 0.

If N contains k digits equal to 8, then the sum of its digits is

$$8k.$$

For divisibility by 3 we require

$$8k \equiv 0 \pmod 3.$$

Since $8 \equiv 2 \pmod 3$, this implies

$$2k \equiv 0 \pmod 3.$$

Thus k must be a multiple of 3.

The smallest possibility is $k = 3$, giving

$$N = 8880.$$

Hence

$$n = \frac{8880}{15} = 592.$$

Therefore

$$\boxed{592}.$$

Remark. *If k denotes the number of digits equal to 8 in N, the condition $8k \equiv 0 \pmod 3$ implies that k must be a multiple of 3.*

Problem 5. *On a table there are 100 tokens. Two players take turns removing tokens from the table. At each turn a player may remove $5, 6, 7, 8, 9,$ or 10 tokens.*

The player who removes the last token wins.

Determine which player has a winning strategy and describe the strategy.

(Combinatorial Game Theory)

Solution. *The first player has a winning strategy.*

The first player begins by removing 10 *tokens, leaving*

$$100 - 10 = 90$$

tokens on the table.

Now suppose the second player removes n *tokens where*

$$5 \leq n \leq 10.$$

The first player responds by removing

$$15 - n$$

tokens.

Thus in every pair of consecutive moves the total number of tokens removed is

$$n + (15 - n) = 15.$$

Since after the first move there are 90 *tokens remaining and*

$$90 = 6 \cdot 15,$$

the number of tokens remaining will decrease as

$$90, 75, 60, 45, 30, 15, 0.$$

Therefore the first player always completes each block of 15 *tokens and will take the last token. Hence the first player wins.*

Problem 6. *Let* f *be a function from the positive integers to the positive integers such that*

$$f(n+1) > f(n)$$

and

$$f(f(n)) = 3n$$

for all positive integers n*.*

Find $f(100)$*.*

(Functional Equations)

Solution. *We will show that for any integers* $k \geq 0$ *and* $0 \leq m < 3^k$*,*

$$f(3^k + m) = 2 \cdot 3^k + m.$$

Since $f(n+1) > f(n)$*, the function is strictly increasing. Hence for any positive integers* n *and* m *we have*

$$f(n+m) \geq f(n) + m.$$

From the functional equation we obtain

$$f(f(1)) = 3.$$

If $f(1) > 2$, *then*

$$3 = f(f(1)) \geq f(f(1) - 1) + 1 \geq f(2) + 1 \geq f(1) + 2,$$

which is impossible.
If $f(1) = 1$, *then*

$$f(f(1)) = f(1) = 1,$$

again a contradiction.
Therefore

$$f(1) = 2.$$

It follows that

$$f(2) = f(f(1)) = 3.$$

We now prove by induction that for every integer $k \geq 0$

$$f(3^k) = 2 \cdot 3^k$$

and

$$f(2 \cdot 3^k) = 3^{k+1}.$$

For $k = 0$ *this is already true since* $f(1) = 2$ *and* $f(2) = 3$.
Assume the statement holds for some k. *Then*

$$f(3^{k+1}) = f(f(2 \cdot 3^k)) = 3 \cdot 2 \cdot 3^k = 2 \cdot 3^{k+1}.$$

Similarly,

$$f(2 \cdot 3^{k+1}) = f(f(3^{k+1})) = 3 \cdot 3^{k+1} = 3^{k+2}.$$

This completes the induction.
Now consider the values

$$f(3^k + m), \qquad m = 0, 1, \ldots, 3^k - 1.$$

These values form an increasing sequence of 3^k *integers contained in the interval*

$$[2 \cdot 3^k, 3 \cdot 3^k - 1].$$

Since this interval contains exactly 3^k *integers, the sequence must fill the entire interval. Hence*

$$f(3^k + m) = 2 \cdot 3^k + m.$$

Finally, we write

$$100 = 3^4 + 19$$

since $3^4 = 81$. *Therefore*

$$f(100) = f(3^4 + 19) = 2 \cdot 3^4 + 19 = 162 + 19 = 181.$$

Thus

$$\boxed{f(100) = 181}.$$